

Guolin Yang

Robotics & World Models, Model-Based RL and Planning

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Profile

PhD researcher in the Robotics and AI group at the University of Manchester, specialising in world models, model-based reinforcement learning, predictive planning, and safe control for autonomous systems. My research focuses on learning compact representations of dynamic, interactive environments and using action-conditioned learned dynamics for multi-step prediction and online planning through MPPI and related sampling-based control methods.

Experienced in Python, C++, PyTorch, CARLA, and closed-loop simulation, with additional automotive engineering experience at Lotus Technology and NIO. My background combines learned world models, planning and decision-making, physical-system dynamics, simulation, and control for real-world autonomous systems.

Research Highlights

- **Interaction-Aware World Models:** Developed a recurrent action-conditioned latent world model for dynamic multi-agent environments, jointly modelling vehicle-pedestrian interaction, future collision risk, reward, and value for sampling-based predictive planning.
- **Use World Models for Predictive Planning:** Built CNN-based BEV and occupancy world models following a latent predictive-control architecture. Spatial observations are encoded into compact latent states and propagated through action-conditioned learned dynamics for multi-step prediction and MPPI planning.

Core Technical Skills

- **World Models & Machine Learning:** Model-Based Reinforcement Learning, Learned Dynamics, Latent World Models, Representation Learning, BEV, Occupancy Network, Reinforcement Learning, Multi-Agent Interaction Modelling
- **Planning & Control:** Model Predictive Path Integral (MPPI) control, Model Predictive Control (MPC), Sampling-Based Planning, Motion Planning, Control Barrier Functions, Risk-Aware Control, Vehicle Dynamics and Control
- **Robotics & Simulation:** CARLA, IPG CarMaker, Closed-Loop Simulation, Autonomous Navigation, Scenario-Based Evaluation, Physical-System Modelling
- **Programming & ML:** Python, C++, PyTorch, MATLAB/Simulink, SciPy, Git, Docker
- **Automotive Systems:** Vector CANoe, CAN Bus, DBC, CAN-Signal Analysis, Diagnostic Analysis
- **Languages:** Chinese (Native), English (Professional), German (A1)

Professional & Research Experience

PhD Researcher – University of Manchester, UK

01/2023 – Present

- Developed world-model-based planning methods for autonomous systems, learning compact latent dynamics of interactive environments and using them for online multi-step decision-making through MPPI-based predictive control.
- Developed an interaction-aware model-based reinforcement learning framework combining latent-state representations, recurrent action-conditioned dynamics, pedestrian-interaction modelling, collision-risk prediction, reward/value estimation, and sampling-based planning.
- Built CNN-based BEV and occupancy world-model variants using TD-MPC-style latent predictive-control architectures, enabling multi-step latent rollout and planning directly from spatial scene representations without explicit image-space reconstruction.
- Integrated learned world models with MPPI trajectory optimisation, allowing candidate control sequences to be evaluated through predicted future system evolution rather than purely reactive policy execution.
- Developed safety-aware predictive-control methods combining sampling-based MPC, Control Barrier Functions, and risk-sensitive optimisation for decision-making under uncertain dynamic-agent behaviour; reduced collisions by up to 65.7% relative to the nominal controller.
- Designed a computationally efficient multi-obstacle safety filter using discrete-time Control Barrier Functions, reducing collision rate from 7.8% to 3.8% across 500 evaluated scenarios with a 9.30 ms runtime.
- Implemented and benchmarked autonomous-navigation agents in Python and PyTorch using custom simulation environments and CARLA, comparing world-model-based, reinforcement-learning, and optimisation-based planning methods.
- Conducted systematic scenario-based evaluation across dynamic multi-agent environments, analysing generalisation, collision risk, minimum clearance, computational latency, and out-of-distribution robustness.
- Supervised Master's dissertation projects in autonomous systems and control, defining technical objectives and providing guidance on modelling, simulation, implementation, and experimental evaluation.

Graduate Teaching Assistant – University of Manchester, UK

09/2023 – Present

- Supported teaching and assessment in Robotic Systems, Applied Mechanics & Industrial Robotics, Digital Control, and Model Predictive Control.

ADAS Engineer – Lotus Technology, Hangzhou, China**07/2022 – 12/2022**

- Supported development of an ADAS speed-limit information feature integrating high-definition map and camera-derived inputs through algorithm-level information fusion and HMI presentation.
- Developed C++ software functions supporting ADAS feature logic and vehicle-signal interfaces.
- Used Vector CANoe for CAN-signal monitoring, CANoe trace analysis, diagnostic-message analysis, functional debugging, and ADAS system integration.
- Investigated feature and integration issues through vehicle-network and diagnostic analysis, and supported functional validation and defect tracking using Jira.
- Conducted DFMEA for the ADAS domain controller, assessing failure modes, severity, occurrence, and detection, and defining potential causes and mitigation actions for assisted-steering, automated-driving, and HMI functions.

BMS Software Intern – NIO, Shanghai, China**06/2021 – 09/2021**

- Developed a three-point high-voltage insulation-detection algorithm in MATLAB/Simulink and supported implementation using model-based development and code-generation workflows.
- Worked with CAN database (DBC) files and developed a user interface to support BMS signal monitoring, data inspection, and system-level debugging.
- Used Vector CANoe to analyse CAN signals and chamber, bench, and vehicle-level test data for software validation and fault investigation.
- Conducted environmental-chamber testing to investigate PTC insulation faults under high-temperature conditions, analysing physical-system behaviour and diagnostic signals to support root-cause identification.

High Voltage Lead – Hunan University Formula Student Electric Team**03/2017 – 12/2019**

- Led system-level concept design and integration of the high-voltage architecture for an electric Formula Student race car.
- Used MATLAB and AVL CRUISE for powertrain-system modelling, performance evaluation, and vehicle-level design decisions.
- Applied DFMEA during high-voltage-system development to identify potential failure modes, define mitigation actions, and support electrical and safety compliance.
- Led lightweight structural design of the HV battery system, achieving a 21% mass reduction while maintaining performance and safety requirements.

Education

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|---|-------------------------------------|-----------------------|
| • PhD in Electrical Engineering | University of Manchester, UK | Jan. 2023 – 2026 |
| Research focus: World models, model-based reinforcement learning, autonomous driving, predictive planning, and safety-critical control.
Supervisor: Prof. Guido Herrmann Co-supervisor: Prof. Alessandra Parisio | | |
| • MSc in Advanced Control and Systems Engineering | University of Manchester, UK | Sep. 2020 – Jun. 2022 |
| Thesis: <i>A Hierarchical Forecasting Model of Pedestrian Crossing Behaviour for Autonomous Vehicles</i> . Research included pedestrian prediction, autonomous-driving decision-making, and control-system modelling. | | |
| • BEng in Vehicle Engineering | Hunan University, China | Sep. 2016 – Jun. 2020 |
| Thesis: <i>Motion Planning and Dynamics Control of Autonomous Racing Car</i> . Developed a four-wheel vehicle dynamics model with Pacejka tyre-force modelling, implemented MPC-based trajectory planning in MATLAB, and evaluated closed-loop trajectory tracking in IPG CarMaker. | | |

Selected Publications

- [1] **Yang, G.**, Chen, A. S., Parisio, A., & Herrmann, G. (2026). A Safety Framework for Uncertainty-Aware Model Predictive Path Integral Control in Shared-Space Navigation. *Proc. IEEE Conference on Decision and Control (CDC)*. Accepted.
- [2] **Yang, G.**, Chen, A. S., Parisio, A., & Herrmann, G. (2026). PIP-CN: Prediction-Integrated Planning for Crowd Navigation. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Accepted.
- [3] **Yang, G.**, Chen, A. S., Parisio, A., & Herrmann, G. (2026). Risk-Bounded Predictive Control for Automated Vehicles in Shared-Space Areas. *International Symposium on Advanced Vehicle Control (AVEC)*. Accepted.
- [4] **Yang, G.**, Lopez Pulgarin, E. J., & Herrmann, G. (2024). A hierarchical forecasting model of pedestrian crossing behavior for autonomous vehicle. *IEEE Access*, 12, 9025-9037.
- [5] **Yang, G.**, & Chai, T. (2020). Time-optimal trajectory planning for autonomous race car. *China Society of Automotive Engineering Annual Conference*.

Honours & Awards

- Full PhD Scholarship, University of Manchester
- Research Excellence Award and Top Student Award, University of Manchester, 2022
- Overall Champion, Formula Student Electric China, 2019